

Batch Creation Automation Tool (KW & Oculus)

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Technology: Python, Pandas

Objective

The Batch Creation Automation Tool automates the preparation of operational batch files from raw CSV datasets. It reduces manual effort by validating input data, applying business rules, standardizing outputs, and generating upload-ready files for different workflows.

Business Problem

The previous process required analysts to manually:

- Download raw CSV files
- Clean product titles
- Remove duplicate records
- Extract values from nested JSON fields
- Generate marketplace-specific Amazon URLs
- Format data according to upload templates
- Save files using standardized naming conventions

This process was repetitive, time-consuming, and susceptible to manual errors.

Proposed Solution

Developed a Python-based automation that:

- Detects available CSV files
 - Allows users to select processing workflow
 - Validates mandatory fields
 - Cleans and standardizes product data
 - Removes duplicate records
 - Generates marketplace-specific URLs
 - Produces upload-ready output files
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Batch Creation Automation Workflow



Solution Components

Component	Responsibility
File Detection	Identifies all available CSV files
Workflow Selection	Supports KW and Oculus processing
Marketplace Mapping	Maps marketplace codes to Amazon domains
Data Validation	Removes invalid or incomplete records
Data Transformation	Extracts JSON values and cleans product titles
Output Generation	Creates standardized upload-ready CSV files

Key Business Rules

- Validate mandatory fields before processing.
 - Remove duplicate ASINs while retaining the first occurrence.
 - Normalize product titles by removing unsupported characters.
 - Extract required values from JSON fields.
 - Generate marketplace-specific Amazon product URLs.
 - Create separate output files for KW and Oculus workflows.
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Error Handling

The automation validates common operational issues including:

- Missing input files
- Unsupported marketplace codes
- Invalid file selections
- Character encoding differences
- Missing mandatory fields

Appropriate error messages are displayed to guide the user.

Benefits

- Reduced manual processing effort

- Improved data consistency
 - Standardized output format
 - Elimination of duplicate records
 - Faster batch preparation
 - Simplified operational workflow
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Conclusion

The Batch Creation Automation Tool provides a modular and maintainable solution for preparing operational batch files. By automating data validation, transformation, and output generation, it improves efficiency, reduces manual effort, and ensures standardized processing across supported workflows.